Unit 5 Human Population and the Environment

- Population growth and Population explosion
- Environment and human health.
- Family Welfare Programme.
- Women and Child Welfare.
- HIV/AIDS
- Human Rights and Value Education.
- Role of Information Technology in Environment.

Characteristics of popualtion

SHIV NADAR UNIVERSITY CHENNAI

- **Population:** Group of individuals of the same species inhabiting the same area simultaneously.
- **Demography:** scientific study of human population, dealing with
  - Changes in population size (growth or decline)
  - Composition of the population (age groups, sex ratio)
  - Distribution of population

- **Characteristics of the population**

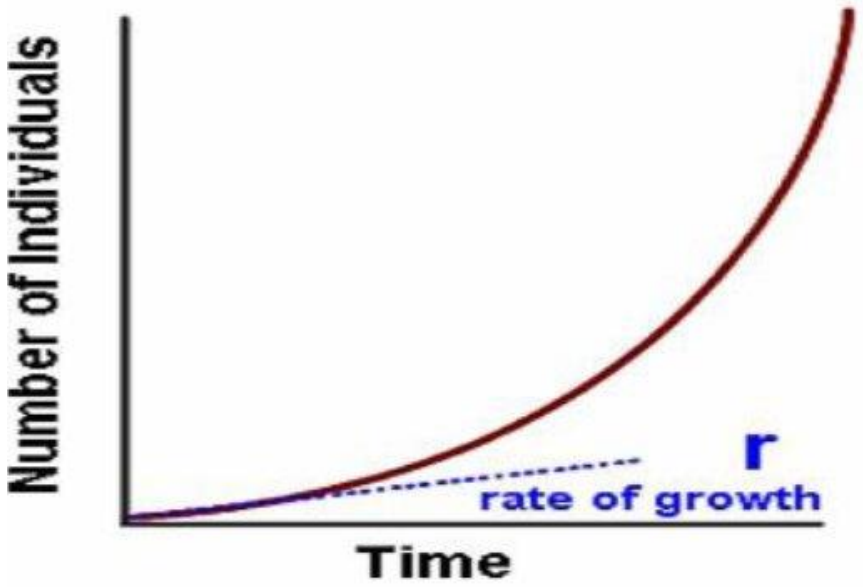
1. **Population growth:** Refers to how the number of individuals in a population increases (or decreases) with time
   1. Exponential
   2. Logistics

Exponential growth- 'J' shaped curve for less developed countries(LDC)

- If a population has a constant birth rate through time and is never limited by food or disease, the birth rate alone controls the population.
- Eg: growth in bacteria, some insects, etc. (not for the human population).

$$P = P_o e^{rt}$$

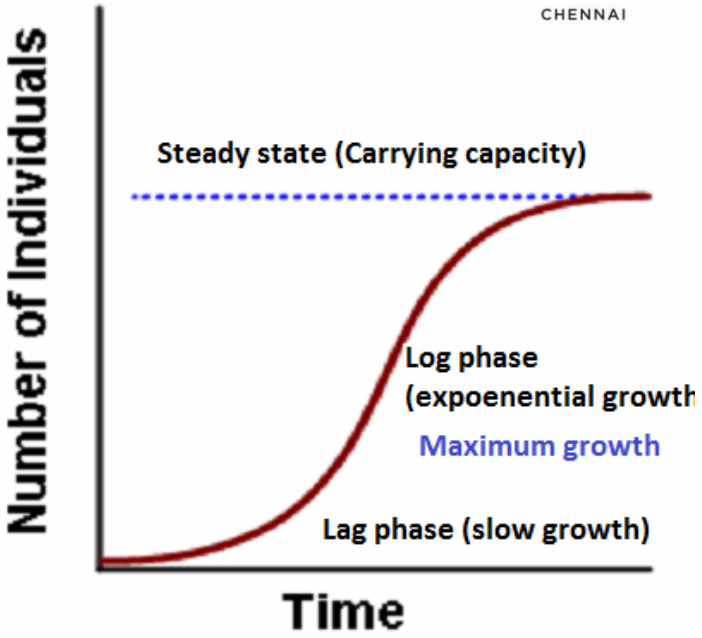
- P: Future population after t years
- Po: Present population
- r: Annual growth rate
- t: Number of years



Logistic Growth: The **curve is 'S' Shaped** form, showing the initial slow growth, followed by steep growth, and finally standardized with less growth.

The logistic curve shows three phases :

- **Lag phase:** little growth
- **Exponential phase:** rapid growth
- **Stationary phase:** stable growth



Zero (population) growth –More developed countries (MDC)

- A condition in which birth plus immigration equals death plus emigration, and population remains constant.
- More developed countries (MDC) show zero population growth.

2. **Population dispersion** It describes how individuals disperse themselves in their habitats.

- **Uniform Dispersion**: Species is evenly distributed about their habitat. This is caused by a species ability to survive anywhere in the habitat. They use the resources found immediately around them and spread out to use all of the available resources • Ex. Arrangement of shrubs in deserts
- **Random Distribution**: Population found randomly about their habitat. Individual have been distributed by chance. Ex: Free floating marine larvae
- **Clumped Dispersion**: Population found in tight clusters, dispersed across a large landscape. This caused when species live together for protection, or natural resources necessary to their survival

3. **Population doubling time**: is the time needed for a population to double its size at a constant annual growth rate.

Population Doubling Time = (ln(2) / Growth Rate)

Where: **ln(2)** is the natural logarithm of 2, approximately equal to 0.6931 and g**rowth Rate** is the annual or periodic growth rate of the population expressed as a decimal.

4. **Total Fertility rate**: Average number of children that would be born to a woman in her lifetime. TFR-1.9 in most developed nations and 4.7 in developing nations or less developed nation. India- 2.6 in 2011

5. **Migration:** The movement of individuals into (immigration) or out of place (emigration)

6. **Female-Male sex ratio:** Sex-ratio is the number of females per thousand males of population.

7. **Life expectancy :** Average age that a newborn infant is expected to attain in a given country.

8. Crude birth rate (CBR): the number of live births in a specific population during a given period (usually a year) per 1,000 people in that population. It is termed "crude" because it does not consider age, gender, or any other demographic characteristics of the population.

Crude Birth Rate = (Number of Births / Total Population) x 1,000

9. Mortality rate: Number of deaths per thousand of individuals in a year.

10.Infant mortality rate: It is the number of infants dying before reaching one year of age, per 1,000 live births.

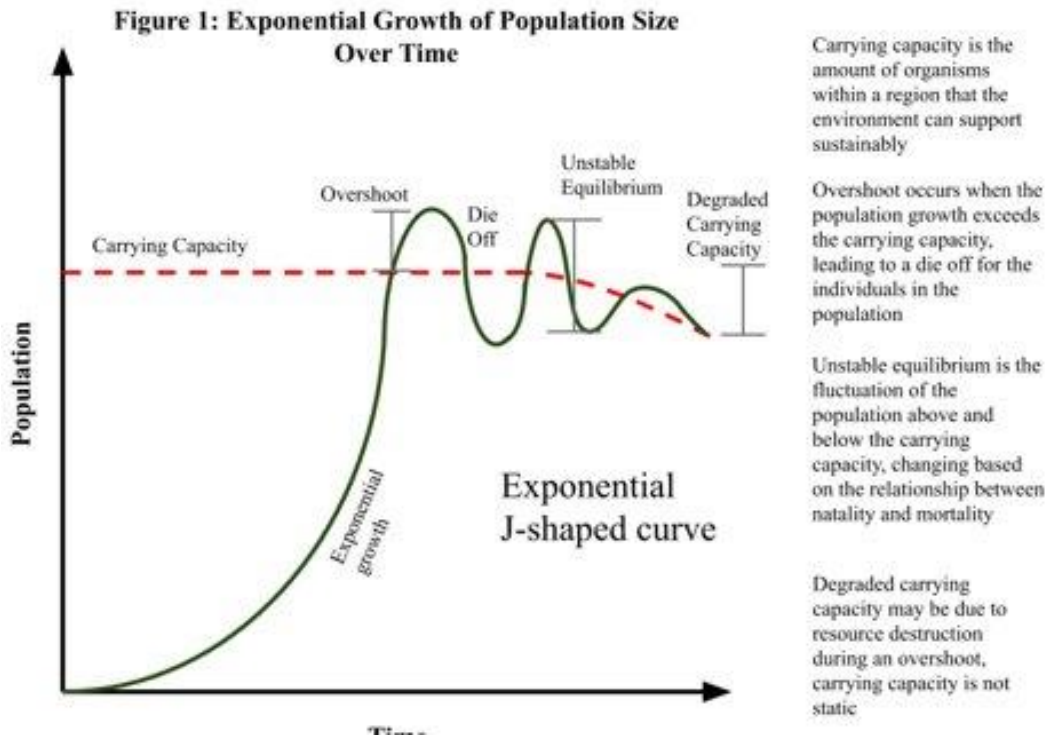
11. Crude death rate (CDR): is a demographic indicator that measures the number of deaths in a specific population during a given period (typically a year) per 1,000 people in that population.

12. Environmental Resistance: Any factor in the environment limiting carrying capacity.

Raw Material and energy availability
Waste Accumulation and Disposal
Organism Interactions

Carrying capacity: Number of individuals of a species that can be indefinitely sustained in a given area without harming the habitat.

- It depends and varies with the habitat.
- If growth exceeds carrying capacity, environmental resistance lowers population size, and the population enters the death phase.
- Species will become threatened, endangered, or extinct.

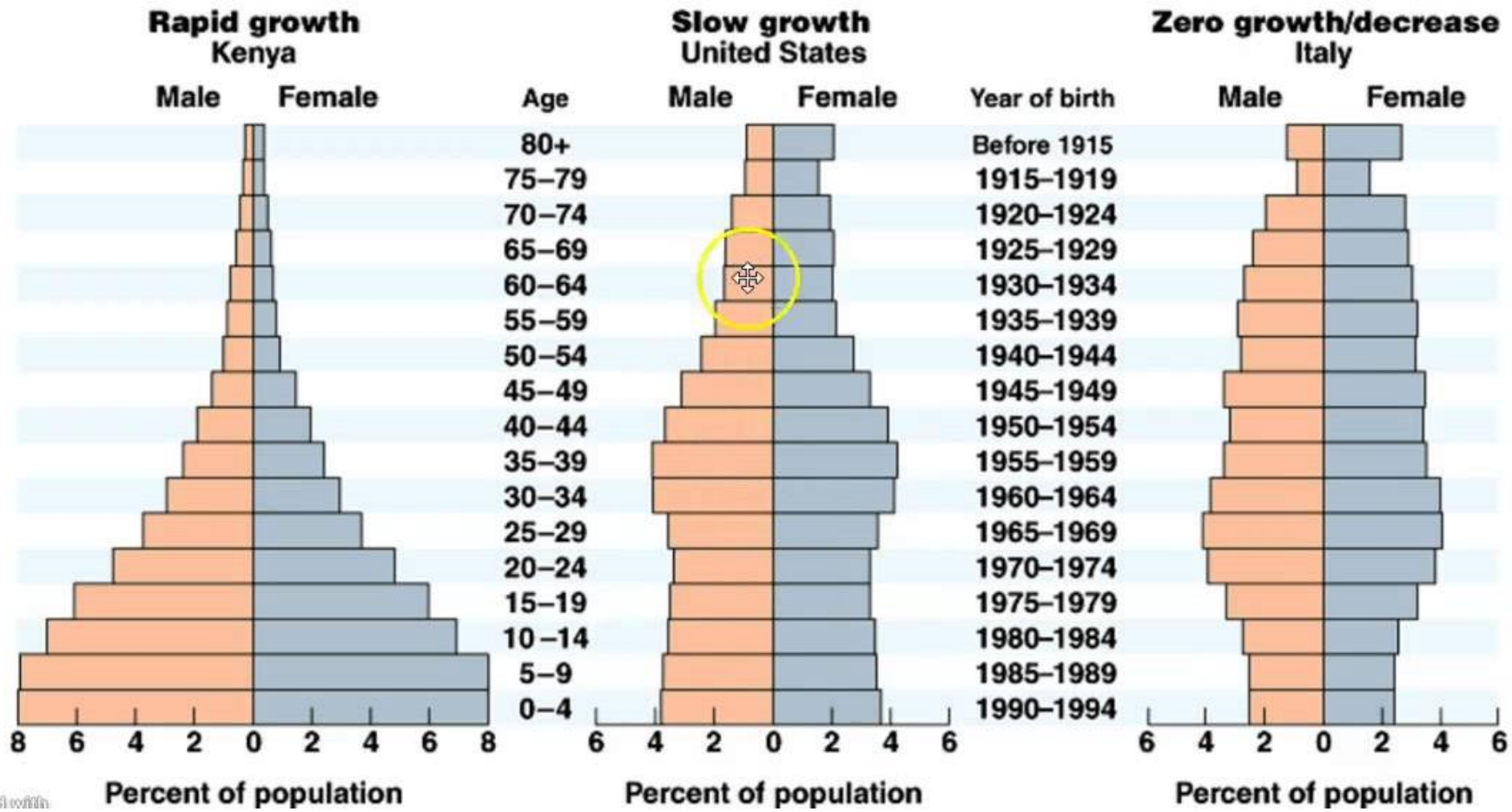


Population Pyramid

SHIV NADAR UNIVERSITY CHENNAI

- **Population pyramid: Represents the age and sex structure of a population graphically.**
- **The shape of the pyramid reflects the characteristics of a population.**
- **Variation in population can be easily represented with the help of a population pyramid.**

- Graphs are on their sides with the axis in the middle.
- Population plotted on the X-axis and age on the Y-axis.
- The number of males is shown on the left and females on the right in the five-year age groups.
- Pre-productive (0-14 yrs.), reproductive (15-44yrs), and post-productive (45 and above)
- Help in understanding
  - Birth and Death Rate
  - Life expectancy
  - Sex ratio
  - Fertility Rate and Infant Mortality Rate
  - Can relate to social & economic problems of the nation



Pyramid

High Fertility ii. High Morality iii. High proportion of children (young age) iv. Moderate Growth Rate. • Developing

Bell

Declining Fertility ii. Declining Mortality iii. Moderate Growth Rate iv. Aging population

Urn

Low fertility ii. Low Mortality iii. Ceasing Growth Rate iv. Very old Population

Population Explosion

SHIV NADAR UNIVERSITY CHENNAI

Population Explosion: A rapid and significant increase in the size of a population caused by such factors as a sudden decline in infant mortality, death rate, or an increase in life expectancy.

REASONS:

- Better health care facilities
- Increased food productivity and distribution:
- Green revolution, the use of scientific methods in agriculture resulted in increased food production, which could feed millions
- Illiteracy and lack of awareness about family planning concepts
- Believes and cultural reasons
- Eradication/control of diseases:
- Government and other agencies funding:

Theories of Population

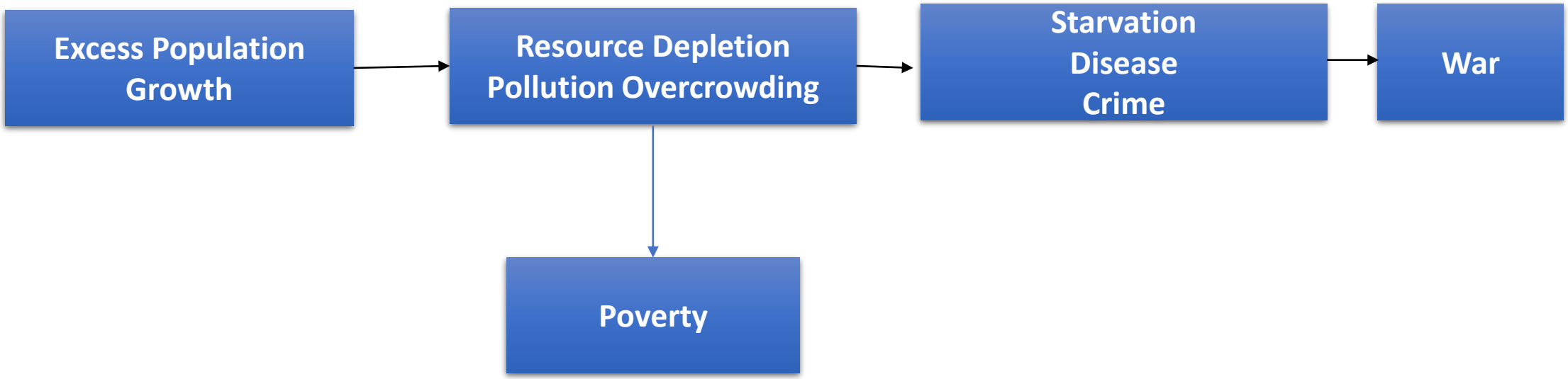
SHIV NADAR UNIVERSITY CHENNAI

Malthusian (Thomas Malthus) Theory :

- “The Principles of Population” hypothesized that there is a relationship between economic development and population growth.
- Population was increasing faster than food production, causing Global Starvation
- Populations tend to grow in **geometric** progression
- Doubling in size every 'n' years, depending on the population growth rate
- Food supplies can, at best, grow in **arithmetic** progression.
- **Population:** 1, 2, 4, 8, 16, 32……..
- **Food: 1, 2, 3, 4, 5…….**

Consequences:

- Population growth will be limited by food availability
- Resulting in Starvation, poverty, disease, crime, misery, and a reduction in population growth.



Marxian (Karl Marx) Theory of population

- Karl Marx contradicted Malthus's view. Marx's view on population is that there is no problem such as overpopulation.
- Social exploitation and oppression of the less privileged people leads to poverty.
- His view was that the problem was caused due to capitalism
- Capitalists facilitate the unequal distribution of food and resources
- Population growth is a symptom rather than a root cause of poverty, resource depletion, pollution, and other social evils.

Remedial Measures:

- Equity and social justice for all
- Allowing everyone to enjoy a good standard of living is a need that can help in achieving a stabilized global population

Environment and Health

SHIV NADAR UNIVERSITY CHENNAI

Health: a state of complete physical, mental and social well-being and not merely the absence of diseases or infirmity.

1. **Infectious diseases**: Unhygienic conditions of environment forms the breeding grounds for various deadly diseases causing organism like virus, bacteria, vectors etc.
    1. **Water-borne(Polluted water):** Cholera, Dysentery, Amoebiosis, Hepatitis
    2. **Air-borne(Polluted air**): Asthma, Bronchitis, Pneumonia, Tuberculosis
    3. **Food-borne(Food-poisoning**): Cholera, Dysentery
    4. **Vector-borne**: Malaria, Typhoid, Filaria
    5. **Animal-borne**: Plague

2. **Noise:** Hearing impairment, hypertension, heart diseases etc.
3. **Radiation:** Cosmic and ultraviolet rays
4. **Diet:** Contamination or undernutrition or malnutrition
5. **Settlement:** Improper settlement and poor physical environment

SHIV NADAR UNIVERSITY CHENNAI

1. Anthropogenic Disposed Chemicals

- **Hazardous**: ignitability, corrosivity, reactivity, or toxicity
- **Toxic Chemicals(Toxins):** kill cells and can serious health concerns
  - *Carcinogenic*: Cause cancer Eg: aromatic hydrocarbons like benzene
  - *Mutagenic:* affect genetic material (DNA) in cells Eg: Polycyclic aromatic hydrocarbon, benzene, sodium azide.
  - *Teratogenic/Embryotoxins*: Cause abnormalities during embryonic growth and development. Eg: various pesticides, heavy metals
  - *Neurotoxins*: Affect nervous system Eg: Lead
- **Pesticide and heavy metal contamination**

2. Occupational Hazards

- **Dust: Lung diseases Silicosis:** Dust contaminating free silica or silicon dioxide-reduces life expectance
- **Asbestosis**: Finest fibres of asbestos find access to respiratory tract leading to respiratory problems

SHIV NADAR UNIVERSITY CHENNAI

HIV/AIDS:

- **HIV:** Human Immunodeficiency Virus is a virus that causes AIDS (Acquired Immunodeficiency Syndrome),
- A health condition in which a person is affected by a series of diseases because of poor immunity.
- **AIDS:** Acquired Immunodeficiency Syndrome is a health condition that results from the deficiency in the body's immunity following HIV infection.

Transmission of HIV:

- Contaminated Fluids
- Contaminated Fluids
- Unprotected sex
- Maternal-fetal transmission
- Improperly sterilized hospital tools

SHIV NADAR UNIVERSITY CHENNAI

Mechanism of HIV infection

- HIV is a retrovirus that damages the immune system by destroying lymphocytes (white blood cells), and helper T cells. Development of HIV
- **Four Stages** of infection
- ***Initial Infection-destruction of helper T cells*** or lymphocytes
- ***Asymptotic or incubation period***
- ***AIDS-related complex (ARC)***
- ***Final stage- Fully infected AIDS patients*** receiving different kinds of infection .

SHIV NADAR UNIVERSITY CHENNAI

How is HIV diagnosed?

- ELISA (Enzyme-Linked Immunosorbent Assay), Immunofluorescent Assay (IFA).
- Western Blot (WB)
- Saliva and Urine test
- Polymerase Chain Reaction (PCR)

Treatment against HIV and AIDS

- **HAART (Highly Active Antiretroviral Therapy)**

Prevention

- Global education about disease
- Use of screened blood samples for disposable infections.
- Protected Sex, monogamy or abstinence from sex
- The risk of HIV transmission from a pregnant woman to her baby is significantly reduced if she takes treatment during pregnancy
- It is necessary to treat STD as soon as you suspect infection

National AIDS Research Institute (NARI)

Women and Child Welfare

SHIV NADAR UNIVERSITY CHENNAI

Special laws concerning women

- The Prohibition of Child Marriage Act, 2006
- Special Marriage Act, 1954
- Dowry Prohibition Act, 1961
- Indian Divorce Act, 1969
- Maternity Benefit Act,1861
- Medical Termination of Pregnancy Act,1971
- Sexual Harassment of Women at Workplace (Prevention, Prohibition and Redressal) Act, 2013
- Indecent Representation of Women(Prevention) Act,1986
- National Commission for Women Act, 1990
- Equal Remuneration Act, 1976
- Protection of Women from Domestic Violence Act, 2005

She-Box Portal : provide speedier remedy to women facing sexual harassment at workplace.
Each case goes directly to the central/ state authority concerned having jurisdiction to take action in the matte

Child welfare in India

SHIV NADAR UNIVERSITY CHENNAI

- **Integrated Child Development Scheme: Launched on 2nd October 1975**
- **Objectives of ICDS:**
  - Lay the foundation for proper psychological development of the child
  - Improve nutritional & health status of children 0-6 years
  - Reduce incidence of mortality, morbidity, malnutrition and school drop-outs
  - Enhance the capability of the mother and family to look after the health, nutritional and development needs of the child.
  - Achieve effective coordination of policy and implementation among various departments

Cradle Baby Scheme

Legislative measures

- Child Labour (Prohibition & Regulation Act) (1986)
- The Child Marriage Restrain Act, 1929
- Balwadi Nutrition programmes –(1970-71)
- Child labour eradication scheme(1984)
- Mid day Meal scheme for school children (1995) -

Protection of Children from Sexual Offences Act (POSCO)

SHIV NADAR UNIVERSITY CHENNAI

The Act defines a child as any person below eighteen years of age, and regards the best interests and well-being of the child as being of paramount importance at every stage, to ensure the healthy physical, emotional, intellectual and social development of the child.

- It defines different forms of sexual abuse all of which are punishable under law.
- People who traffic children for sexual purposes are also punishable under the provisions relating to abetment in the Act.
- The Act prescribes stringent punishment graded as per the gravity of the offence, with a maximum term of rigorous imprisonment for life, and fine

Family Welfare Program

SHIV NADAR UNIVERSITY CHENNAI

Family planning is the planning of when to have children, how many children to have and the use of birth control and other techniques to implement such plans.

Family planning measures:

- Use of contraceptives
- Condom (For male's use)
- Diaphragm (For female's use)
- Intrauterine Contraceptive Device (IUD)
- Norplant
- Jellies, creams, and foam
- Oral contraceptive
- Sterilization
- Vasectomy
- Tubectomy

Human Rights in India

SHIV NADAR UNIVERSITY CHENNAI

1. **Right to equality:** All are treated equal irrespective of religion, race, caste, gender or place of birth. All have equality of opportunity in matters of employment.
2. **Right to freedom:** This right includes freedom for speech and expression.
3. **Right against exploitation**: prohibits all forms of forced labour, child labour and trafficing
4. **Right to freedom of religion:**
5. **Cultural and Educational rights**: Right of minorities.
6. **Right to constitutional remedies**: Enforcement of Fundamental Rights.
7. **Right to life:** Right to live with human dignity, including right to education, health, shelter and basic amnesties that the state shall provide

Values and Ethics

SHIV NADAR UNIVERSITY CHENNAI

- **Values** are things (an object, a person or an idea which are considered important or a way of behaving), serving as guiding principles which influence our behavior and determine whether or not such behavior seems worthwhile.

- **Value vs Education**

- **Ethics:** A set of standards or value system by which human actions are determined as right or wrong.
- Morals: Conforming to an established set of codes or excepted notions of right and wrong.

Framework for Ethical Decision Making: STAR

1. **STOP**. – Examine the situation with who, what, where, and when?
2. **THINK.** – Every action has a consequence, both intended and unintended.
3. **ACT**. – Choose the best alternative and take appropriate action.
4. **REVIEW**. – Once a course of action is chosen and implemented, then review it.

Role of information technology in environment

SHIV NADAR UNIVERSITY CHENNAI

i) **ENVIS – Environmental Information System**: decentralized system with a network of Centres ensuring environmental information collection, storage, retrieval and dissemination to all concerned.

- Generates network of database on areas like pollution control, clean technologies, remote sensing, coastal ecology, biodiversity,
- OBJECTIVES OF ENVIS
  1. To build up a repository and dissemination centre in Environmental Science and Engineering
  2. To support and promote research, development and innovation in environmental information technology.
  3. c) To promote exchange of information amongst developing countries.

ii) **NIMS – National Management Information System:** building the information base on a continuous basis on resources devoted to scientific and technological activities for policy planning in the country.

iii. **RIS – Remote Sensing Information System:** collection of data by sensors not in di

- Satellite imageries provide the actual information about various physical and biological resources and their degradation in a digital form through remote sensing.
- Providing correct, reliable and verifiable information about forest cover, conservation efforts
- Provides information of ozone layer depletion, smog, approach of monsoon etc.

iv. **GIS – Geographical Information System:** capture, store, manipulate, analyze, manage, and present all types of geographical data.

i) Different thematic maps containing digital information on a number of aspects like water resources, industrial growth, road network, soil type, forest land, grass land etc. are superimposed in a layered form incomputer using software.
ii) For interpretation of polluted zones, degraded lands, emission sources etc.
iii) Zoning Atlas is prepared by GIS to locate suitable areas for industrial growth
iv) Resource mapping, environmental conservation, management, planning, Coral reef mapping etc.
v) Helps in disaster Management and relief operations

v. **WWW– World Wide Web**

vi. **Computer based instruments for environment**

Role of information technology in health

SHIV NADAR UNIVERSITY CHENNAI

Combination of IT and health is **bioinformatics.**

- Electronic health record (EHR)
- Automated dispensing machines (ADMs)
- Picture archiving and communications system (PACS)
- Teleconferencing
- Endoscopy and Laparoscopy
- Instruments in health care
- Mobile blood banks

Environmental Impact Analysis/Assessment

Environmental Impact Assessment (EIA) is a process of evaluating the likely environmental impacts of a proposed project or development, taking into account inter-related socio-economic, cultural, and human-health impacts, both beneficial and adverse.

EIA would necessarily involve the following stages:

- Screening
- Scoping
- Preparing the Environmental Impact Statement (EIS) or EIA report
- Review of the Environmental Impact Statement (EIS)
- Decision-making
- Monitoring, compliance, enforcement, and environmental auditing.